

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

*I _Nalluri Akhilesh Kumar(192365074) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

N Akhilesh Kumar
Signature of the student:

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 1: Article Summary

(a) Citation, Domain and ML Problem

The complete citation is:

Oh, S. H., Park, J., Lee, S. J., Kang, S., & Mo, J. (2022). *Reinforcement learning-based expanded personalized diabetes treatment recommendation using South Korean electronic health records*. *Expert Systems with Applications*, 206, 117932. <https://doi.org/10.1016/j.eswa.2022.117932>.

The research belongs to the healthcare and precision medicine domain. It addresses the problem of selecting personalized diabetes medication treatments according to the patient's health condition.

The authors focus on the fact that diabetes treatment is not identical for every patient. Patient characteristics such as age, complications, family history, smoking, BMI, blood pressure, fasting plasma glucose and existing medication can affect treatment decisions.

The research uses Reinforcement Learning, specifically a contextual bandits approach, to recommend diabetes medication. The study also compares the proposed method with Q-learning, Thompson Sampling, Lin-UCB, Actor-Critic, DQN and Double-DQN.

(b) Key Objective and Research Gap

150–200 words

The main objective of the article is to develop a more realistic personalized diabetes treatment recommendation system using reinforcement learning and South Korean electronic health records. Existing treatment recommendation approaches often use only a limited number of patient characteristics, such as fasting plasma glucose, which may not adequately represent the complex decision-making process used by doctors. The authors identified a research gap in representing the high-dimensional nature of real clinical decision-making while maintaining a manageable reinforcement learning model. Traditional Markov Decision Process approaches can become difficult to implement when many patient characteristics and possible treatment actions are included because the state-action space becomes very large. To address this limitation, the authors proposed a contextual bandits-based approach that incorporates more detailed patient information. The proposed system considers demographic, medical history, lifestyle and clinical measurements and recommends mono-, dual-, or triple-

drug therapy according to patient conditions. The authors evaluated the proposed approach against several established reinforcement learning algorithms and examined whether the recommendations could improve important diabetes-related health measurements such as fasting plasma glucose, BMI and blood pressure.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 2: Methodology Analysis

(a) ML Techniques Used

The main technique used in the research is Reinforcement Learning, particularly a Contextual Bandits model.

Why Contextual Bandits?

The authors explain that a conventional MDP can become difficult when the number of patient states increases.

The proposed system therefore uses contextual information about the patient to select appropriate treatment actions.

The study represents patient conditions using several variables, including:

- Sex
- Age
- Diabetes complications
- Duration/period of diabetes
- Family history
- Smoking
- Existing medication intake
- BMI
- Blood Pressure
- Fasting Plasma Glucose (FPG)

These variables produce 1,296 possible patient states, while the treatment space contains 15 actions representing combinations of diabetes medications.

Dataset

The study uses electronic health records from the South Korean National Health Insurance Sharing Service (NHISS).

The data includes information such as:

- Medical history
- Patient characteristics
- Medication information
- Laboratory measurements
- Diabetes complications
- FPG
- BMI
- BP

The authors state that the underlying Korean database is updated with approximately one million patient records per year. The exact research dataset is not publicly available because it contains potentially identifying medical information and requires controlled access.

The study received Institutional Review Board approval from Seoul National University of Science and Technology, with IRB number 2021-0030-01.

(b) Mapping to CO4

CO4 Topic	Application in Article
Inductive Learning	The system learns treatment-selection patterns from historical patient records.
Decision Trees	Not the main technique used.
Statistical Learning	Not the primary method.
Reinforcement Learning	Main technique used to learn personalized treatment recommendations.

Therefore, this paper has a direct and strong connection to CO4, particularly the Reinforcement Learning component.

Methodological Strength

A major strength is that the authors use more comprehensive patient states instead of relying only on a single measurement such as blood glucose.

The model considers demographic, lifestyle, medical and clinical information. This makes the representation closer to actual clinical decision-making.

Methodological Limitation

The major limitation is the single-source dataset.

The research uses Korean health records from one national database. The authors themselves state that additional medical databases would be useful for comparing performance across

different races and ethnicities. They also did not include HbA1c, an important diabetes-related measurement.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 3: Results & Critical Evaluation

(a) Key Results

The authors evaluated the proposed contextual bandits model against six reinforcement learning approaches:

1. Q-Learning
2. Thompson Sampling
3. Lin-UCB
4. Actor-Critic
5. Deep Q-Network (DQN)
6. Double-DQN (D-DQN)

The proposed model demonstrated better performance than the compared reinforcement learning methods according to the study's evaluation.

The model also produced clinically meaningful treatment patterns.

For patients older than 45 years, the recommendation for monotherapy decreased from 37% to 17% as patient conditions became more severe, indicating that the model adapted treatment recommendations according to patient state.

The study further reports that patients receiving treatments concordant with model recommendations experienced improvements in diabetes-related measurements, including:

- Fasting Plasma Glucose
- BMI
- Blood Pressure

It also reports delayed occurrence of diabetes-related complications among patients following model-concordant recommendations.

Comparative Table

Model	Type	Role in Study
Contextual Bandits	Proposed RL approach	Proposed personalized recommendation model
Q-Learning	RL	Benchmark
Thompson Sampling	Bandit/RL	Benchmark
Lin-UCB	Contextual Bandit	Benchmark
Actor-Critic	Deep RL	Benchmark
DQN	Deep RL	Benchmark
D-DQN	Deep RL	Benchmark

The important point is that this article does not primarily evaluate itself using conventional classification metrics such as Accuracy, Precision, Recall and F1-Score. Because this is a treatment recommendation problem, the evaluation focuses more on treatment concordance and subsequent clinical outcomes.

(b) Critical Evaluation

Are the Results Realistic?

The results are promising, but they should not be interpreted as proof that the AI can independently prescribe medication.

The model was developed and evaluated using retrospective electronic health records. Therefore, it learns from treatment decisions that were already made by healthcare professionals.

A strong association between model-concordant treatment and better outcomes does not automatically prove that the model itself caused those improvements.

Potential Biases

1. Dataset Bias

The dataset represents patients from the South Korean healthcare system. Therefore, the results may not generalize directly to Indian, European, American or other populations.

2. Treatment Selection Bias

Doctors may have selected treatments based on factors that are not fully represented in the dataset.

3. Missing Variables

The authors specifically identify the absence of HbA1c as a limitation.

4. Historical Bias

The model learns from historical clinical decisions. If historical treatment decisions contain bias, the model may reproduce that bias.

5. Offline Evaluation

The model was not tested through a prospective randomized clinical trial. Therefore, real-world effectiveness still needs further validation.

Additional Experiments That Would Strengthen the Research

I would recommend:

1. Testing the model on datasets from different countries.
2. Including HbA1c and additional clinical variables.
3. Performing external validation.
4. Conducting subgroup analysis by age and sex.
5. Comparing recommendations against independent expert clinicians.
6. Performing prospective clinical trials.
7. Testing model robustness when some patient information is missing.
8. Performing fairness analysis across different demographic groups.

These experiments would improve confidence in the model's generalizability and clinical safety.

(c) Real-World Applicability

The proposed technique could be deployed as a Clinical Decision Support System (CDSS).

Possible applications include:

1. Hospitals

The system could analyze patient records and provide treatment recommendations to doctors.

2. Diabetes Clinics

Doctors could receive personalized suggestions based on the patient's current health state.

3. Electronic Health Record Systems

The RL model could be integrated into an EHR platform to generate recommendations automatically.

4. Telemedicine

Patient information collected remotely could be analyzed to assist clinicians in choosing appropriate treatment options.

5. Population Health Management

Healthcare organizations could use the system to identify treatment patterns associated with better long-term outcomes.

Industry Challenges

Data availability: Large, high-quality longitudinal medical datasets are difficult to obtain.

Privacy: Medical records contain sensitive patient information and require strong security and access controls.

Computational requirements: Deep RL models can require substantial computing resources.

Integration: The AI system must integrate with existing hospital/EHR systems.

Explainability: Doctors need to understand why a treatment was recommended.

Regulatory approval: Medical AI systems may require extensive validation and regulatory approval before clinical deployment.

Patient safety: An incorrect recommendation could potentially cause serious harm.

Therefore, the system should be implemented as doctor-assisted decision support rather than an autonomous prescribing system.

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

Insight 1 – Patient State Representation is Critical

I learned that the effectiveness of a Reinforcement Learning system depends strongly on how the state is represented.

The article does not represent a diabetic patient using only blood glucose. It includes age, sex, BMI, BP, family history, complications, medication and other factors.

Connection to CO4:

This directly relates to MDP state representation in Reinforcement Learning.

A better state representation can allow the agent to make more personalized decisions.

Insight 2 – Large State Spaces Create Computational Challenges

I learned that increasing the number of patient characteristics can make a conventional MDP extremely large.

The study addresses this problem using Contextual Bandits rather than relying only on a conventional MDP representation.

Connection to CO4:

This deepens my understanding of Reinforcement Learning, state-action spaces and decision-making.

It demonstrates that choosing an RL formulation is itself an important engineering decision.

Insight 3 – Historical Data Can Support Personalized Decisions

I learned that Reinforcement Learning can be applied to historical electronic health records rather than only through direct interaction with a live environment.

The model learns relationships between patient states, treatment actions and resulting outcomes.

Connection to CO4:

This connects Reinforcement Learning with data-driven decision-making and policy learning.

It also shows why healthcare RL requires careful evaluation because historical data may contain biases from previous clinical decisions.

(b) Proposed Novel Extension

Multi-Country Fairness-Aware Reinforcement Learning

If I were extending this research, I would develop a fairness-aware personalized diabetes treatment recommendation system using multiple international EHR datasets.

Instead of training and evaluating the model only on South Korean patients, I would combine appropriately harmonized datasets from different populations, such as:

- South Korea
- India
- United States
- European populations

The model would include additional clinically relevant features such as:

- HbA1c
- Fasting glucose
- BMI
- Blood pressure
- Age
- Family history
- Medication history
- Diabetes complications
- Lifestyle information

I would then compare the model's performance across demographic groups and countries.

Why is this feasible?

The proposed system is based on patient-state information and treatment outcomes, so the same RL framework can be adapted to other EHR datasets after appropriate preprocessing and feature harmonization.

Expected Impact

This extension could:

- Improve generalization.
- Reduce population-specific bias.
- Identify differences in treatment effectiveness.
- Improve fairness.
- Increase confidence in real-world deployment.
- Make the system more useful for healthcare systems outside South Korea.

The original authors themselves suggest using additional medical databases to compare different races and ethnicities, making this a logical extension of their work.

Overall Critical Review

The selected article provides a strong practical example of Reinforcement Learning being applied to a real healthcare decision-making problem. Its major contribution is the use of a contextual bandits approach to represent a larger set of patient characteristics while avoiding some of the dimensionality problems associated with conventional MDP-based treatment recommendation. The study is particularly relevant to CO4 because it demonstrates how an AI agent can learn treatment policies from patient states and outcomes.

The use of real-world electronic health records is a major strength because it makes the research more representative of actual healthcare environments. However, the reliance on a single Korean healthcare database, the absence of HbA1c, the retrospective evaluation design, and the lack of prospective clinical validation limit the extent to which the results can be generalized.

Overall, the research demonstrates that Reinforcement Learning has strong potential for personalized medical decision support, but clinical deployment requires additional validation, fairness analysis, explainability, privacy protection and human oversight. The system should

therefore be viewed as a decision-support tool for healthcare professionals rather than an autonomous treatment-prescribing system.

Article Link

DOI: 10.1016/j.eswa.2022.117932

This article is a particularly good choice for your submission because it directly connects to the CO4 topic of Reinforcement Learning and also uses the same diabetes-treatment scenario as your previous assessment.